FLEXIPAY

FlexiPay Wallet

OPEN APIs

Payments Interface Specification Document V 1.6

About this document

Purpose

This document explains how to integrate to SBU FlexiPay wallet through the Webservice interface. FlexiPay wallet uses eConnect ESB is the middleware to connect FlexiPay managed channels e.g., POSs, mobile applications, USSD, portals to wallet and to the 3rd party providers

Objective and Scope

	Transaction
1.	Merchant Payments/Collections
2.	Bulk Payments
3.	Bulk Registration
4.	Funds Transfer/cashouts
5.	Third Party Collection
6.	Bill Payments

Glossary & Acronyms of Terms

Abbreviation	Meaning
APIC	API Connect
ISO	International Standards Organization
MOMO	Mobile Money
C2B	Customer to Business
TCP	Transmission Control Protocol
CA	certificate authority
Dig-Signature	Digital Signature

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Communication Protocol

- A RESTFUL web service over TCP/IP

Message Standard

A customized JSON derivation of the ISO 8583 standard.

Authentication and Security

1. Partner needs to provide an **X-IBM-Client-Id** to the message header. This is got when you create an APP (API key) on our developer portal provided you have an account.
2. Partner needs to provide **X-IBM-Client-Secret** to the message header. This is got when you create an APP (API Secret) on our developer portal provided you have an account.
3. Partner needs to provide **password** to the message header. For this password we will share it with you.
4. Partner needs to provide an **"Authorization Bearer Token"** to the message header. It's a way to grant access to protected resources without revealing sensitive credentials like usernames and passwords.
   a. **Token Expiry and Refresh**: To enhance security, bearer tokens often have an expiration time of 15minutes(900s). After the token expires, the client needs to re-authenticate to obtain a new token.
5. To manage data integrity of request and response of API's
   a. Generate a **CA Signed** Certificate/private key pair for signing and verifying signatures using **sha256 RSA Digital Signature Algorithm of 2048bits**. Note you will share with us the Certificate to configure it on our Server.
   b. Sign requests: Before sending a request to the server, sign the request body with no spaces using the private key. Attach the signature to the request header as **x-signature** so that the server can verify it.
   c. Verify signatures on the server: On the server, will extract the signature from the request header and verify it using the certificate. If the signature is valid, process the request; otherwise, reject it (you will get a 401 unauthorised).

Certificate Requirements

We will require a CA certificate (public key) of a SHA-256 Encryption with RSA-2048 signature
A **certificate authority (CA)** is a trusted organization that issues digital certificates for websites and other entities.
You will use your private Key to sign your requests and generate a signature and send it in headers as x-signature.

Please Note
Self-signed certificates are not allowed
Please generate a certificate under a domain you manage for example **api.companyName.com**

In case you don't have a CA certificate you are to get one from **any CA of your own choice.**

You can also check up on this website below there is 24hrs live chat personnel ready support and guide you on which type of certificate you require, please note remember to ask for guidance for sha256RSA certificate of 2048bits. Link to the website: https://cheapsslweb.com/sectigo-positive-ssl

You could use openssl to generate CSR here is sample command below and then to submit the CSR to CA of your choice to issue you a certificate.

```
openssl req -out testing.csr -new -newkey rsa:2048 -nodes -keyout testing.key
```

How to get the bearer Token

oauth

Type

oauth2

Token URLs

https://gateway.apps.sandbox.stanbicbank.co.ug/ug/oauth2/token

Scopes

Create - This scope will allow consumers to create any resource

var Basic_token = new Buffer(APP-Key + ':' + APP-Secret).toBase64

```
curl --location 'https://gateway.apps.sandbox.stanbicbank.co.ug/ug/oauth2/token' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--header 'Authorization: Basic ' \
--data-urlencode 'grant_type=client_credentials' \
--data-urlencode 'scope=Create'
```

response


```
{
    "token_type": "Bearer",
    "access_token": "AAIgMjRhNmQ3NDE2OTIyODkyMDcwZjFlM2ZmYmVmNDA1NWRQOYa_BJl8fePEMOgZ7Lsc_bDZ6
AZkAZqgEuhXiMUI2X0qweR8C3uOeK-wkXQO8GquAPtyPtEkZGfr9StpG8e9t8JRQziWHxrX63jG7l4X2wuDZBOUZb-
AJvtrp82m08s",
    "scope": "Create",
    "expires_in": 900,
    "consented_on": 1689330157
}
```


Put the token in the headers as Authorization: "**Bearer ${**`access_token`**}**"

How to generate the X-Signature

Python snippet

Python snippet you must sign your request using your Private key and generate the signature and send it as x-signature in headers

```
import json
import re
import base64
from Crypto.Signature import pkcs1_15
from Crypto.Hash import SHA256
import json
import base64
from Crypto.PublicKey import RSA

# Load private key from file
with open("private_key.pem", "rb") as key_file:
    private_key = RSA.import_key(key_file.read())
    print (private_key)


payload =
{"requestId":"239071178528","transactionType":"ABAL","accountId":"9030017144234"}
# remove white spaces from payload
message = json.dumps(payload)
message = re.sub(r'\s+', '', message)

# Convert the message to bytes
message_bytes = message.encode()

# Hash the JSON string using SHA256
hash_obj = SHA256.new(message_bytes)

# Sign the hash using the private key and RSA-PSS padding
signature = pkcs1_15.new(private_key).sign(hash_obj)

# Encode the signature in base64
encoded_signature = base64.b64encode(signature).decode('utf-8')

print (encoded_signature)
print (message)
```

Java snippet

Java snippet you must sign your request using your Private key and generate the signature and send it as x-signature in headers

```
import java.nio.charset.StandardCharsets;
import java.security.KeyFactory;
import java.security.PrivateKey;
import java.security.Signature;
import java.security.spec.PKCS8EncodedKeySpec;
import java.util.Base64;
import java.nio.file.Files;
import java.nio.file.Paths;

public class RRSASignature {
    public static void main(String[] args) throws Exception {
        // Read the private key from a file
        String privateKeyFilePath = "path/to/your/private-key.pem"; // Replace with the
actual file path
        String pkcs8PrivateKey = new
String(Files.readAllBytes(Paths.get(privateKeyFilePath)), StandardCharsets.UTF_8);

        // Original payload even if it has spaces
        String payload = "{\"msisdn\":\"256703\",\"requestId\":\"302211\",\"clientId\":
\"NUAT\",\"merchantCode\": \"853\"}";

        // Remove spaces from payload for consistent processing
        String payloadWithoutSpaces = payload.replaceAll("\\s", "");
        byte[] messageBytes = payloadWithoutSpaces.getBytes(StandardCharsets.UTF_8);

        // Remove header and footer from the private key, and decode it from Base64
        String privateKeyPEM = pkcs8PrivateKey.replace("-----BEGIN PRIVATE KEY-----",
"")
                .replace("-----END PRIVATE KEY-----", "")
                .replaceAll("\\s+", "");
        byte[] privateKeyDecoded = Base64.getDecoder().decode(privateKeyPEM);

        // Generate a private key object from the decoded key
        PKCS8EncodedKeySpec keySpec = new PKCS8EncodedKeySpec(privateKeyDecoded);
        PrivateKey privateKey = KeyFactory.getInstance("RSA").generatePrivate(keySpec);

        // Create a Signature object and initialize it with the private key
        Signature sign = Signature.getInstance("SHA256withRSA");
        sign.initSign(privateKey);

        // Update the Signature object with the message
        sign.update(messageBytes);
```

```
        // Generate the RSA signature
        byte[] signatureBytes = sign.sign();
        String signatureBase64 = Base64.getEncoder().encodeToString(signatureBytes);

        // Print the payloadWithoutSpaces and RSA signature
        System.out.println("Payload: " + payloadWithoutSpaces);
        System.out.println("RSA Signature: " + signatureBase64);
    }
}
```

Node.js snippet

Node.js snippet you must sign your request using your Private key and generate the signature and send it as x-signature in headers

```
const crypto = require('crypto');
const fs = require('fs');

// Function to convert PKCS#8 private key to PKCS#1 private key
function convertPkcs8ToPkcs1(pkcs8PrivateKey) {
  const key = crypto.createPrivateKey(pkcs8PrivateKey);
  return key.export({ type: 'pkcs1', format: 'pem' });
}

// Load the PKCS#8 private key from a file
const pkcs8PrivateKeyFilePath = 'C:/path/pkcs8-private-key.pem'; // Replace with the
actual file path
const pkcs8PrivateKey = fs.readFileSync(pkcs8PrivateKeyFilePath, 'utf-8');

// Convert PKCS#8 private key to PKCS#1 private key
const pkcs1PrivateKey = convertPkcs8ToPkcs1(pkcs8PrivateKey);

const payload = { "name": "doe" };
const message = JSON.stringify(payload);

const sign = crypto.createSign('RSA-SHA256');
sign.update(message);
const signature = sign.sign(pkcs1PrivateKey, 'base64');

console.log('MessageBody:', message);
console.log('RSA Signature:', signature);
```

Headers

Method: **POST**

```
curl --location 'endpoint\
--header 'X-IBM-Client-Id:  \
--header 'X-IBM-Client-Secret:  \
--header 'x-signature:  ' \
--header 'password: ' \
--header 'Content-Type: application/json' \
--header 'Authorization: Bearer ' \
--data ' Payload '
```

Note: The spaces in the payload have to first be removed before signing it, then the payload where the spaces have been removed it's what is sent as the request body with its corresponding Signature generated passed in the headers as x-signature.

Merchant Payments

Method: **POST**

```
 curl --location endpoint' \

--header 'X-IBM-Client-Id: ' \
--header 'X-IBM-Client-Secret: ' \
--header 'x-signature:  ' \
--header 'password: ' \
--header 'Content-Type: application/json' \
--header 'Authorization: Bearer ' \
--data 'payload'
```

API Requests and Responses

1.Merchant Lookup / Validate Merchant

Description: Returns the name and location of a FlexiPay merchant based on the merchant code provide.

Merchant lookup API URL
https://gateway.apps.sandbox.stanbicbank.co.ug/fp/v1.1/validatemerchant

Merchant Lookup Request


```
{
  "msisdn":"256759889380",
  "requestId":"202211141638",
  "merchantCode":"228971",
  "clientId":"HCLUAT"
}
```

Merchant Lookup Response


```
{
  "statusCode":"00",
  "statusDescription":"Completed successfully",
  "merchantName":"KAKANDE ENTERPRISES",
  "merchantLocation":"Metro",
  "flexipayReferenceNumber":"50000000731"
}
```


2. Merchant Payment – From Customer Mobile Money Wallet

Description: Initiates a merchant payment with the customer's Mobile Money Wallet as a source of funds (MTN or Airtel).

Merchant payment API URL

https://gateway.apps.sandbox.stanbicbank.co.ug/fp/v1.1/merchantpayment

Merchant collection/payment Request


```
{
        "msisdn":"256701221117",
        "requestId":"202211141638",
        "merchantCode":"228971",
        "referenceNumber":"10000001",
        "amount":"3000001",
        "sourceSystem":"AIRTEL",
        "narrative":"Lunch Money",
        "clientId":"HCLUAT"
}
```


Merchant collection/payment Response


```
{
  "statusCode": "00",
  "statusDescription": "Your transaction has been received and is being processed,
  "transactionReferenceNumber": "10000001",
  "flexiPayReferenceNumber": "500000007479"
}
```


3. Get Merchant Payment Status (Mobile Money Payments)

Description: Fetches the status of a merchant payment with the mobile money wallet as source of funds.

Get transaction status API URL

https://gateway.apps.sandbox.stanbicbank.co.ug/fp/v1.1/merchantpaymentstatus

Get Transaction Status Request


```
{
        "requestId":"202211141638",
        "clientId":"FLEXUAT"
```

}

Get Transaction Status Response

{

"statusCode": "00",
"statusDescription": "successfully Completed",
"transactionReferenceNumber": "202211141638",
"flexipayReferenceNumber": "500000005591"
}

4. Initiate Approval – From Customer FlexiPay Wallet

Description: Initiates a merchant payment with the customer's FlexiPay Wallet as a source of funds.

Initiate approval API URL

https://gateway.apps.sandbox.stanbicbank.co.ug/fp/v1.1/initiateapproval

Initiate approval Request

{
"msisdn":"256759889380",
"requestId":"202211141600",
"merchantCode":"228971",
"referenceNumber":"10000001",
"amount":"10000",
"sourceSystem":"FLEXIPAY",
"narrative":"Lunch Money",
"clientId":" FLEXUAT "
}

Initiate approval Response

{
"statusCode": "00",
"statusDescription": "Your transaction has been received and is being processed,
"transactionReferenceNumber": "10000001",
"flexiPayReferenceNumber": "500000007481"
}

5. Get Customer Approval Status (FlexiPay Payments)

Description: Fetches the status of a merchant payment approval with the FlexiPay wallet as source of funds.

Note – This applies to customer approvals

Get approval status API URL

https://gateway.apps.sandbox.stanbicbank.co.ug/fp/v1.0/getApprovalStatus

Get Transaction Status Request

{
"clientId":" FLEXUAT ",
"requestId":"202211141600"
}

Get Transaction Status Response


```
{
   "statusCode": "00",
   "statusDescription": "Successfully Completed",
   "clientId": "HCLUAT",
   "requestId": "202211141600",
   "flexipayReferenceNumber": "500000007482"
}
```


Error Response

When the Authorization bearer token is not passed in the headers or when it has expired.


```
{
   "httpCode": "401",
   "httpMessage": "Unauthorized",
   "moreInformation": "Cannot pass the security checks that are required by the target API or operation, Enable debug headers for more details."
}
```


When your Signature doesn’t match the request body sent.


```
{
   "httpCode": 401,
   "httpMessage": "Unauthorized",
   "moreInformation": "Signature Validation Failed."
}
```


When you pass wrong X-IBM-Client-Id and Secret


```
{
   "httpCode": "403",
   "httpMessage": "Forbidden",
   "moreInformation": "Multiple client ids found."
}
```

Merchant Payments Call Back From FlexiPay

For the merchant payment call back the partner is expected conform to our security standards.

Security Requirements - OAuth2 and message signing

Sample Curl to get an **authorization Token.**

```
curl --location 'tokn-url' \
--header 'Content-Type: application/json' \
--header 'Authorization: Basic ${base64EncodedToken}' \
--data '{
   "grant_type": "client_credentials",
   "scope": "Create"
}'
```

- curl: This is the command-line tool used for making HTTP requests.
- --location 'token-url': This part specifies the URL where the token will be requested. Replace 'token-url' with the actual URL where the token should be obtained.
- --header 'Content-Type: application/json': This sets the HTTP request header to indicate that the data being sent is in JSON format.
- --header 'Authorization: Basic ${base64EncodedToken}': Here, the Authorization header is used for basic authentication. ${base64EncodedToken} should be replaced with the actual base64-encoded token.
- --data ‘{“grant_type": "client_credentials", "scope": "Create" }': This part contains the data to be sent with the request. It specifies the grant_type as "client_credentials" and the scope as "Create," indicating the type of access being requested.

On receiving the Authorization token, we will then send this token in the headers while calling the rest of your endpoints.

Authorization: `Bearer ${Access_Token} ` and x-signature in headers.

Message signing

To manage data integrity of request and response of API’s, we request you implement message signing on your end.

- We to Generate a public key/private key pair for signing and verifying signatures using sha256 RSA Digital Signature Algorithm. Note We will share with you our public key to configure it on your Server.

- Sign payload request: We will sign the payload request Before sending the payload request to your server. We will then attach the generated signature to the request header as x-signature so that your server can verify it.

Merchant Payment call back from FlexiPay

1. Merchant payment call back.

merchant payment call back to partner

```
{
  "statusCode": "00",
  "statusDescription": "Completed successfully",
  "transactionReferenceNumber": "18389303058",
  "flexiPayReferenceNumber": "500076003433",
  "customerNumber": "256707286362",
  "narrative": "food payment",
  "amount": "500076003433"
}
```

Acknowlegdement response from the Partner

```
{
  "statusCode": "00",
  "statusDescription": "successfully"
}
```

Request and Response Message Fields

Merchant Request Messages Fields:

Parameter name	Description	Required	Format
msisdn	MNO mobile number, payment number	Mandatory	String
requestId	Transaction reference	Mandatory	String
clientId	Client uniquely assigned code	Mandatory	String
merchantCode	Merchant unique code	Mandatory	String

referenceNumber	For referenced merchants	Optional	String
amount	Transactional amount	Mandatory	String
narrative	Transaction narration/reason for payment	Mandatory	String
sourceSystem	Source of value. Can be FLEXIPAY, AIRTEL or MTN. If source is FLEXIPAY, msisdn must be the	Mandatory	String

Merchant Response Messages Fields:

Parameter name	Description	Required	Format
statusDescription	Transaction Status	Mandatory	String
requestId	Transaction ID from partner	Mandatory	String
flexipayReferenceNumber	Transaction ID from partner	Mandatory	String
statusCode	Transaction status code	Mandatory	String

Bulk Registration

1. Bulk Registration

Description: This API enables to onboard of non FlexiPay wallets to the FlexiPay wallets.

```
curl --location ' \
--header 'X-IBM-Client-Id:  \
--header 'X-IBM-Client-Secret:  \
--header 'x-signature:  ' \
--header 'password:  ' \
--header 'Content-Type: application/json' \
--header 'Authorization: Bearer \
--data 'payload'
```

Bulk Registration API URL

https://gateway.apps.sandbox.stanbicbank.co.ug/fp/v1.0/bulkRegistration

Bulk Registration Request

Note – For this request you will need to include your callback URL. A callback will be sent from FlexiPay to this URL once the registration has been completed suceesfully.


```
{
   "requestId": "1211368701",
   "requestTime": "29/04/1997 12:35:08",
   "NumberOfRecords": "1",
   "clientId": "FLEXIUAT",
   "narrative": "Wallet onboarding ",
   "customerData": [
      {
         "cardNumber": "018375565",
         "mobileNumber": "256702345678",
         "dob": "29/04/1997",
         "firstName": "John",
         "lastName": "Byenkya",
         "secondName": "",
         "nin": "CM9702610872HG",
         "gender": "MALE",
         "occupation": "Developer"
      }
   ]
}
```


Bulk Registration Response


```
{
   "Status": "00",
   "StatusMessage": "Registration details has been received successfully."
}
```


Error Response

When the Authorization bearer token is not passed in the headers or when it has expired.


```
{
   "httpCode": "401",
   "httpMessage": "Unauthorized",
   "moreInformation": "Cannot pass the security checks that are required by the target API or operation, Enable debug headers for more details."
}
```


When your Signature doesn't match the request body sent.


```
{
   "httpCode": 401,
```

"httpMessage": "Unauthorized",
"moreInformation": "Signature Validation Failed."
}

When you pass wrong X-IBM-Client-Id and Secret

{
"httpCode": "403",
"httpMessage": "Forbidden",
"moreInformation": "Multiple client ids found."
}

Bulk Registration **Request Messages Fields:**

Parameter name	Description	Required	Format
requestTime	Transaction date and time in the format DD/MM/YYYY HH:MM:SS	Optional	String
requestId	Transaction reference	Mandatory	String
clientId	Client uniquely assigned code	Mandatory	String
numberOfRecords	Total number of records contained under customer data	Mandatory	String
narrative	Transaction narration/reason for registration payment	Mandatory	String
customerData	An array of json objects containing customer data	Mandatory	String
mobileNumber	Customer MSISDN (Phone number) starting with 256 with only numbers	Mandatory	String
firstName	Customer first name as registered at MNO	Mandatory	String
secondName	Customer second(middle) name as registered at MNO	Mandatory, empty string when the customer doesn't have	String
lastName	Customer last name as registered at MNO	Mandatory	String

nin	Uganda national identification number	Mandatory	String
gender	customer gender	Mandatory	String/Integer
occupation	Customer occupation (source of income)	Mandatory	String/integer

Bulk Registration **Response Messages Fields:**

Parameter name	Description	Required	Format
StatusMessage	Description of transaction status	Mandatory	String
Status	Transaction status code	Mandatory	String

Bulk Payments

1. Bulk Payments

Description: Initiates a payment to the bulk FlexiPay wallets specified.

```
curl --location ' \
--header 'X-IBM-Client-Id:  \
--header 'X-IBM-Client-Secret:  \
--header 'x-signature:  ' \
--header 'password:  ' \
--header 'Content-Type: application/json' \
--header 'Authorization: Bearer \
--data 'payload'
```

Bulk payments API URL
https://gateway.apps.sandbox.stanbicbank.co.ug/fp/v1.1/bulkpayment

Bulk payment Request

Note – For this request you will need to include your callback URL. A callback will be sent from FlexiPay to this URL once the transaction has been posted on the specified wallets to notify you of the payment status


```
{"requestId":"20210706033645",
"requestTime":"2023/04/19 14:26:08",
"numberOfRecords":"3",
"totalAmount":"30000",
"clientId":"HCLUAT",
"narrative":"Allowances",
"callbackURL":" Partner callback URL here",
"customer_data":
    [
        {
        "mobileNumber":"256773234110",
        "firstName":"Bernard",
        "secondName":"Mabelle Soto",
        "lastName":"Cheress",
        "amount":"10000",
        "paymentNarration":"Weekly Allowance"},
        {
        "mobileNumber":"256759556327",
        "firstName":"Kipkirui",
        "secondName":"",
        "lastName":"Bett",
        "amount":"10000",
        "paymentNarration":"Weekly Allowance"},
        {
        "mobileNumber":"256721411331",
        "firstName":"Kakande",
        "secondName":"MW",
        "lastName":"Sabiti",
        "amount":"10000",
        "paymentNarration":"Weekly Allowance"}]
}
```


Bulk payment Response


```
{
   "statusCode": "00",
   "callbackURL": null,
   "statusDescription": "Payment initiated successfully."
}
```

2.Bulk payment callback to partner

Description: – This callback will be sent from FlexiPay to your specified callback URL once the transaction has been posted on the specified wallets, to notify you of the payment status.


```
{
"requestId":"20210706033645",
"requestTime":"2022/10/29 14:26:08",
"customerData":
        [{"mobileNumber":"256773234110",
        "statusCode":"00","amount":"100000",
        "statusMessage":"Completed successfully"},
        {"mobileNumber":"256759556327",
        "statusCode":"57","amount":"100000",
        "statusMessage":"Provided name mismatch with registered name on Flexipay"},
        {"mobileNumber":"256721411331",
        "statusCode":"00","amount":"100000",
        "statusMessage":"Completed successfully"}]
}
```


Bulk payment callback response from partner


```
{
   "Status": "00",
   "StatusMessage": "True"
}
```


Error Response

When the Authorization bearer token is not passed in the headers or when it has expired.


```
{
   "httpCode": "401",
   "httpMessage": "Unauthorized",
   "moreInformation": "Cannot pass the security checks that are required by the target API or operation, Enable debug headers for more details."
}
```


When your Signature doesn't match the request body sent.


```
{
```

"httpCode": 401,
"httpMessage": "Unauthorized",
"moreInformation": "Signature Validation Failed."
}

When you pass wrong X-IBM-Client-Id and Secret

{
"httpCode": "403",
"httpMessage": "Forbidden",
"moreInformation": "Multiple client ids found."
}

Bulk Payment Request Messages Fields:

Parameter name	Description	Required	Format
Request_Time	Transaction date and time in the format YYYY/MM/DD HH:MM:SS	Mandatory	String
RequestID	Transaction reference	Mandatory	String
Client_ID	Client uniquely assigned code	Mandatory	String
Reference_no	For referenced merchants	Mandatory	String
Amount	Transactional amount	Mandatory	String
Narrative	Transaction narration/reason for batch payment	Mandatory	String
customer_data	An array of json objects containing customer data	Mandatory	String
mobile_number	Customer MSISDN (Phone number) starting with 256 with only numbers	Mandatory	String
first_name	Customer first name as registered at MNO	Mandatory	String

second_name	Customer second(middle) name as registered at MNO	Mandatory, empty string when the customer doesn't have	String
last_name	Customer last name as registered at MNO	Mandatory	String
payment_narration	Reason for payment (Individual payment description	Mandatory	String
amount	Amount to pay individual customer	Mandatory	String/Integer
no_of_records	Total number of records contained under customer data	Mandatory	String/integer
Total_amount	Sum of all amount under customer data	Mandatory	String/Integer

Bulk Payment Response Messages Fields:

Parameter name	Description	Required	Format
StatusMessage	Description of transaction status	Mandatory	String
Status	Transaction status code	Mandatory	String

Funds Transfer

```
curl --location 'endpoint' \
--header 'X-IBM-Client-Id: ' \
--header 'X-IBM-Client-Secret: ' \
--header 'x-signature:  ' \
--header 'password:  ' \
--header 'Content-Type: application/json' \
--header 'Authorization: Bearer ' \
--data 'payload'
```

API Requests and Responses

1. Get Wallet Details

Description: Fetches the details of a FlexiPay Wallet number specified.

Get wallet details API URL

https://gateway.apps.sandbox.stanbicbank.co.ug/fp/v1.1/validatewalletdetails

Get wallet details Request


```
{
        "requestId":"202211141638",
        "msisdn":"256759889380",
        "clientId":"HCLUAT"
}
```


Get wallet details Response


```
{
  "statusCode": "00",
  "statusDescription": "Successful",
  "walletName": "Samantha Piwan",
  "currency": "UG",
  "allowCredit": "N",
  "allowDebit": "N"
}
```


2.Transfer to FlexiPay Wallet (inbound Remittance)

Description: Posts funds to a FlexiPay wallet.

Wallet Payment API URL

https://gateway.apps.sandbox.stanbicbank.co.ug/fp/v1.1/walletpayment

Wallet payment Request


```
{
 "requestId": "30004560012",
 "clientId": "CLIENT",
 "narrative": "Concert Tickets",
 "originatorName": " Namanya",
 "originatorSurname": "Kevin",
 "originatorCountryOfResidence": "Uganda",
 "originatorDateOfBirth": "10/03/1957",
```

"originatorNationality": "Ugandan",
"beneficiaryAccount": "256702221139",
"beneficiaryName": "Anna",
"beneficiarySurname": "Namuli",
"beneficiaryCountryOfResidence": "Uganda",
"mobileNumber": "256702221139",
"amount": "500",
"network": "WALLET",
"destination": "WALLET"
}

Wallet payment response

{
"statusCode": "00",
"statusDescription": "Successful",
"requestId": "30004560012",
"flexipayId": "500000007499"
}

3. Transfer to FlexiPay Wallet - Local

Description: Posts funds to a FlexiPay wallet.

Wallet Payment API URL

https://gateway.apps.sandbox.stanbicbank.co.ug/fp-domestic/v1.0/walletpayment

Wallet payment Request

{
"requestId": "30004560012",
"clientId": "CLIENT",
"narrative": "Concert Tickets",
"beneficiaryAccount": "256702221139",
"beneficiaryName": "Namuli Anna",
"mobileNumber": "256702221139",
"amount": "500",
"network": "WALLET",

"destination": "WALLET"
}

Wallet payment response

{
"statusCode": "00",
"statusDescription": "Successful",
"requestId": "30004560012",
"flexipayId": "500000007499"
}

4. Transfer to a FlexiPay Merchant - Local

Description: Posts funds to a FlexiPay business merchant code.

Wallet Payment API URL

https://gateway.apps.sandbox.stanbicbank.co.ug/fp/v1.0/transfer-to-merchant

Transfer to a merchant Request

{
"requestId": "30004560012",
"clientId": "CLIENT",
"narrative": "Concert Tickets",
"beneficiaryAccount": "223487",
"beneficiaryName": "Namuli and Sons Enterprise",
"mobileNumber": "256702221139",
"amount": "500",
"network": "WALLET",
"destination": "WALLET"
}

Transfer to a merchant response

{
"statusCode": "00",
"statusDescription": "Successful",
"requestId": "30004560012",
"flexipayId": "500000007499"
}

5.Transfer to Mobile Money Wallet (inbound Remittance)

Description: Posts funds to a mobile money wallet.

Note: For MNO number lookup destination field will be sent as KYC and the phones names will be returned if the beneficiary number is valid at MNO

Sample field "destination": "KYC"

Wallet Payment API URL

https://gateway.apps.sandbox.stanbicbank.co.ug/fp/v1.1/mobiletransfer

MNO transfer Request


```
{
 "requestId": " TEST98720007",
 "clientId": "CLIENT",
 "narrative": "Concert Tickets",
 "originatorName": " Namanya",
 "originatorSurname": "Kevin",
 "originatorCountryOfResidence": "Uganda",
 "originatorDateOfBirth": "10/03/1957",
 "originatorNationality": "Ugandan",
 "beneficiaryAccount": "256772899790",
 "beneficiaryName": "Anna",
 "beneficiarySurname": "Namuli",
 "beneficiaryCountryOfResidence": "Uganda",
 "mobileNumber": "256772899768",
 "amount": "500",
 "network": "MTN",
 "destination": "MNO"
}
```


MNO Transfer response


```
{
   "statusCode": "00",
   "statusDescription": "Successful",
   "requestId": " TEST98720007",
   "flexipayId": "500000007499"
}
```


MNO Lookup response


```
{
   "requestId": " TEST98720008",
```

"StatusCode": "00",
"flexipayId": "500023056821",
"statusDescription": "NAMULI ANNA"
}

6.Transfer to Mobile Money Wallet - Local

Description: Posts funds to a mobile money wallet.

Note: For MNO number lookup destination field will be sent as KYC and the phones names will be returned if the beneficiary number is valid at MNO

Sample field "destination": "KYC"

Wallet Payment API URL

https://gateway.apps.sandbox.stanbicbank.co.ug/fp-domestic/v1.0/mobiletransfer

MNO transfer Request

{
"requestId": " TEST98720001",
"clientId": "CLIENT",
"narrative": "Concert Tickets",
"beneficiaryAccount": "256772899790",
"beneficiaryName": "Namuli Anna",
"mobileNumber": "256772899768",
"amount": "500",
"network": "MTN",
"destination": "MNO"
}

MNO Transfer response

{
"statusCode": "00",
"statusDescription": "Successful",
"requestId": " TEST98720001",
"flexipayId": "500000007499"
}

MNO Lookup response

{
"requestId": " TEST98720002",
"StatusCode": "00",

"flexipayId": "500023056821",
"statusDescription": "NAMULI ANNA"
}

7. Transfer to Bank (inbound Remittance)

Description: Posts funds to a Stanbic or external bank account wallet.

Transfer to Bank URL

https://gateway.apps.sandbox.stanbicbank.co.ug/fp/v1.1/banktransfer

Bank Transfer Request

{
"requestId":"TPSC0006547465",
"clientId":"TECHPAY",
"bankSortCode":"SBICUGKXXXX",
"bankName":"Stanbic Bank Uganda",
"originatorName":"Matthew",
" originatorSurname ":"Davidson",
"originatorCountryOfResidence":"Ireland",
"originatorDateOfBirth":"10/04/1968",
"originatorNationality":"Irish",
"mobileNumber":"256701221117",
"narrative":"Guitar Lessons",
"amount":"10000",
"beneficiaryAccount":"9030012345678",
"beneficiaryName":"Alan",
"beneficiarySurname":"Mukisa",
"beneficiaryCountryOfResidence":"Uganda",
"network":"BANK","currency":"UGX",
"destination":"BANK"
}

Bank Transfer Response

{
"requestId": "TPSC0006547465",
"flexipayId": "500000010837",
"statusCode": "00",
"statusDescription": "Your transaction has been received and is being processed"
}

8. Transfer to Bank to local

Description: Posts funds to a Stanbic or external bank account wallet.

Transfer to Bank URL

https://gateway.apps.sandbox.stanbicbank.co.ug/fp-domestic/v1.0/banktransfer

Bank Transfer Request


```
{
    "requestId":"TPSC0006547465",
    "clientId":"TECHPAY",
    "bankSortCode":"SBICUGKXXXX",
    "bankName":"Stanbic Bank Uganda",
    "mobileNumber":"256701221117",
    "narrative":"Guitar Lessons",
    "amount":"10000",
    "beneficiaryAccount":"9030012345678",
    "beneficiaryName":"Mukisa Alan",
    "network":"BANK","currency":"UGX",
    "destination":"BANK"
}
```


Bank Transfer Response


```
{
  "requestId": "TPSC0006547465",
  "flexipayId": "500000010837",
  "statusCode": "00",
  "statusDescription": "Your transaction has been received and is being processed"
}
```


9.Get Account Name

Description: Fetches the account name of a account number specified.

Get Account Name API URL

https://gateway.apps.sandbox.stanbicbank.co.ug/fp/v1.0/GetAccountName

Get account name request


```
{
    "requestId": "300089081150",
    "clientId": "FLEXUAT",
    "beneficiaryAccount": "9030012345678"
}
```

Get account name response


```
{
   "statusDescription": "Request Successful.",
   "requestId": "300089081150",
   " name": "AKIDING ALICE",
   "statusCode": "00"
}
```


10. Get Transaction Status

Description: Fetches the status of a transaction

Note – This applies to customer approvals

Get approval status API URL

https://gateway.apps.sandbox.stanbicbank.co.ug/fp/v1.0/GetTransactionStatus

Get Transaction Status Request


```
{
        "clientId":"HCLUAT",
        "requestId":"20210706033645"
}
```


Get Transaction Status Response


```
{
   "statusCode": "57",
   "statusDescription": "No payment received request ID",
   "clientId": "HCLUAT",
   "requestId": "20210706033645",
   "flexipayReferenceNumber": ""
}
```


Error Response

When the Authorization bearer token is not passed in the headers or when it has expired.


```
{
   "httpCode": "401",
   "httpMessage": "Unauthorized",
   "moreInformation": "Cannot pass the security checks that are required by the target API or operation, Enable debug headers for more details."
}
```


When your Signature doesn't match the request body sent.


```
{
   "httpCode": 401,
```

"httpMessage": "Unauthorized",
"moreInformation": "Signature Validation Failed."
}

When you pass wrong X-IBM-Client-Id and Secret

{
"httpCode": "403",
"httpMessage": "Forbidden",
"moreInformation": "Multiple client ids found."
}

Fund Transfer Request Messages Fields:

Parameter name	Description	Required	Format
clientId	Client uniquely assigned code	Mandatory	String
requestId	Transaction ID from partner	Mandatory	string
requestTime	Timestamp of transaction initiation	Mandatory	String
bankSortCode	Sort code of recipient bank	Mandatory – Applies to transfer to bank requests only	String
bankName	Name code of recipient bank	Mandatory	String
originatorName	First name of the person sending funds	Mandatory	String
originatorSurname	Surname of the person sending funds	Mandatory	String
originatorCountryOfResidence	Country of residence of sender of funds	Mandatory	String
originatorDateOfBirth	Date of birth of residence of sender of funds	Mandatory	String
originatorNationality	Nationality of sender of funds	Mandatory	String
mobileNumber	Mobile number that will receive transactional SMS	Mandatory	String
narrative	Transaction narration	Mandatory	String
amount	Transaction amount	Mandatory	String
beneficiaryAccount	Recipient account number (Bank number, mobile money number or wallet	Mandatory	String

	account number, Business Partner Merchant code)		
beneficiaryName	Recipient first name	Mandatory	String
beneficiarySurname	Recipient surname	Mandatory	String
beneficiaryCountryOfResidence	Country of residence of the recipient of funds	Mandatory	String
network	MNO Network of recipients of funds	Mandatory	String
destination	Destination where funds will terminate	Mandatory	String

Funds Transfer Response Messages Fields:

Parameter name	Description	Required	Format
StatusDesc	Transaction Status	Mandatory	String
requestId	Transaction ID from partner	Mandatory	String
flexipayId	FlexiPay reference number	Mandatory	String
StatusCode	Transaction status code	Mandatory	String

Third Party Collections through FlexiPay

2. Customer Validation

Validation request to partner

```
{
   "request_reference": "E00112545201",
   "customer_number": " AB123456",
   "transtype": "presentment"
}
```

Validation response from partner

```
{
   "status_code": "00",
   "status_desc": "Completed successfully",
   "customer_name": "John Doe",
```

```
   "pending_balance": "5500",
   "payment_mod": "PART"
}
```

3. Customer Payment

Payment request to partner

```
{

   "request_reference": "EC10000343443",
   "customer_number": " AB123456",
   "customer_name": "John Doe",
   "transtype":"payment",
   "narrative": "Payment from customer AB123456",
   "reason_for_payment": "loan_repayment",
   "location": "Kampala",
   "amount": "182000"
}
```

Payment response from partner

```
{
   "status_code": "00",
   "status_desc": "Completed successfully",s
   "external_reference": "18389303058",
   "flexipay_ref_num": "EC10000343443"
}
```

Bill Payments

Request Authentication

```
curl --location 'endpoint \
--header 'X-IBM-Client-Id:  \
--header 'X-IBM-Client-Secret:  \
--header 'x-signature:  ' \
--header 'password:  \
--header 'Content-Type: application/json' \
--header 'Authorization: Bearer ' \
--data 'payload'
```

API Requests and Responses

1. Customer validation at biller

Biller payment validation API URL:
https://gateway.apps.sandbox.stanbicbank.co.ug/fp/v1.0/billercustomervalidation

Customer validation request to FlexiPay


```
{
   "ServiceID": "1001",
   "ClientId":"FLEXUAT",
   "requestReference": "000112545201",
   "customerId": "042578934007",
   "Transtype": "presentment",
   "customerMobile": "256700123456",
   "Location": "Kampala"
}
```


Customer validation response from FlexiPay


```
{
   "StatusCode": "00",
   "StatusDesc": "Completed successfully",
   "customer_name": "John Mark",
   "Amount": "32000",
   "FlexipayRefNum": "EC10000343443",
   "processid": "Flexi3434585455650022",
   "defaultPackage": "GOTVSUPA",
   "uedlid": "1202106221106065757",
   "TVcustomernumber": "259883734",
   "schoolCode": "100"
}
```


2. Bill Payment

Biller payment API URL:
https://gateway.apps.sandbox.stanbicbank.co.ug/fp/v1.0/billpayment

Bill payment request to FlexiPay


```
{
    "serviceID": "1001",
    " clientId": "FLEXUAT",
    "requestReference": "000112545201",
    "narrative": "Bill payment for DSTV",
    "customerId": "11111",
    "customerName": "Amos Okello",
    "tin": "58955",
    "uedlid": "1202106221106065757",
    "processid":"Flexi3434585455650022",
    "productCode": "GOtvPLS",
    "location": "Kampala",
    "customerMobile": "254711940525",
    "amount": "200",
    "schoolCode": "100"
}
```

Bill payment response from FlexiPay


```
{
   "statusCode": "00",
   "statusDesc": "Completed successfully",
   "requestReference": "000112545290",
   "uedlid": "1202106221106065757",
   "processid": "Flexi3434585455650022",
   "electricityToken": "62253816533927452555",
   "electricityUnits": "16.5",
   "flexipayRefNum": "EC10000343487",
   "receiptNumber": "0000000009083138"
}
```


UEDCL amount validation request to Flexipay


```
{
   "serviceID": "1008",
   "clientId": "FLEXUAT",
   "customerId": "11111",
   "transtype": "presentment",
   "operation": "amountvalidation",
   "requestReference": "18389303058",
   "processid": "Flexi3434585455650022",
   "uedlid": "1202106221106065757"
}
```


UEDCL amount validation response from Flexipay


```
{
   "statusCode": "00",
   "statusDesc": "Completed successfully",
   "customer_name": "Test Test",
   "Amount": "5500",
   "FlexipayRefNum": "EC10000343443"
   "uedlid": "1202106221106065757"
}
```


Fetch TV packages request to Flexipay


```
{
   "serviceID": "1005",
   "clientId": "FLEXUAT",
   "customerId": "11111",
   "transtype": "presentment",
   "operation": "fetchtvPackages",
```

```
   "requestReference": "18389303058"
}
```

Fetch TV packages response from Flexipay

```
{
   "statusCode": "00",
   "statusDesc": "Completed successfully",
   "customer_name": "Test Test",
   "amount": "5500",
   "flexipayRefNum": "EC10000343443"
   "tvpackages": "GOTV^GOTV VALUE BOUQUET|16000~GOtvPLS^GOtv Plus
Bouquet|26000~GOLITE^GOTV LITE BOUQUET|65000~GOTVMAX^GOTVMAX - Gotv Max
Bouquet|49000~GOTVSUPA^GOtv Supa Bouquet|55000~"
}
```

3. Check Status of Bill payment

Description: This API is used to check status of Bill payment to confirm the transaction.

Check Status Bill payment API URL:
https://gateway.apps.sandbox.stanbicbank.co.ug/fp/v1.0/billpaymentstatus

Bill payment Check Status Request

```
    {
    "clientId":"FlexUAT",
    "requestReference":"12873653413231
    }
```

Bill payment Check Status Response

```
       {
    "statusCode": "00",
    "statusDescription": "Dear customer,Your transaction was successful",
    "transactionReferenceNumber": "12873653413231",
    "flexipayReferenceNumber": "500000143496",
    "token": "",
    "units": ""
}
```

4. Check aggregator wallet balance

Check aggregator balance API URL:

https://gateway.apps.sandbox.stanbicbank.co.ug/fp/v1.0/getWalletBalance

Balance Check Request

```
{
   "clientId": "FLEXUAT",
   "requestReference": "18389303058"
}
```

Balance Check Response

```
{
   "statusCode": "00",
   "statusDesc": "Completed successfully",
   "availableBal": "550000",
   "actualBal": "550000",
   "flexipayRefNum": "3000000001236"
}
```

Error Response

When the Authorization bearer token is not passed in the headers or when it has expired.

```
{
   "httpCode": "401",
   "httpMessage": "Unauthorized",
   "moreInformation": "Cannot pass the security checks that are required by the target API or operation, Enable debug headers for more details."
}
```

When your Signature doesn't match the request body sent.

```
{
   "httpCode": 401,
   "httpMessage": "Unauthorized",
   "moreInformation": "Signature Validation Failed."
}
```

When you pass wrong X-IBM-Client-Id and Secret

```
{
```

"httpCode": "403",
"httpMessage": "Forbidden",
"moreInformation": "Multiple client ids found."
}

Biller Service IDs

Service ID	Biller	Flow
1001	UMEMEYAKA	This service has 2 set of operations: 1. Presentment/Validation 2. Payment
1002	UMEMEPOST_PAID	This service has 2 set of operations: 1. Presentment/Validation 2. Payment
1003	NWSC	This service has 2 set of operations: 1. Presentment/Validation 2. Payment
1005	DSTV	This service has 3 set of operations: 1. Presentment/Validation 2. Fetch TV packages 3. Payment
1006	GOTV	This service has 3 set of operations: 1. Presentment/Validation 2. Fetch TV packages 3. Payment
1007	SCHOOLPAY	This service has 2 set of operations: 1. Presentment/Validation 2. Payment
1008	UEDCL	This service has 3 set of operations: 1. Presentment/Validation 2. Amount Validation 3. Payment
1009	NSSF	This service has 2 set of operations: 1. Presentment/Validation 2.Payment

,

Request and Response Message Fields

Header Message Fields

```
curl --location 'endpoint' \
--header 'X-IBM-Client-Id: ' \
--header 'X-IBM-Client-Secret:  ' \
--header 'x-signature:  ' \
--header 'password:  ' \
--header 'Content-Type: application/json' \
--header 'Authorization: Bearer ' \
--data 'payload'
```

Bill payment Request Messages Fields:

Parameter name	Description	Required	Format
ServiceID	Code that identifies the payment item being Paid for.e.g. 1001 for Umemeyaka Bill presentment.	Mandatory	String
ClientId	Id of Client making the request	Mandatory	String
requestReference	The request reference that uniquely Identifies the transaction.	Mandatory	String
Narrative	Transaction narration	Optional	String
Reason for payment	Caters to the different deposit categories a partner may have on their system e.g savings deposit vs loan repayment	Optional	String
Amount	Transactional amount	Mandatory	String
customerId	The customer unique identifier being Validated or paid for e.g. meter number, Account No, PRN, TRN, SmartCardNumber, phone number in the case of airtime top-up etc.	Mandatory	String
customerName	The name of the UMEME customer making the payment **N.B – Only for UMEME payments**	Optional	String

Tin	Taxpayer Identification Number. This is returned upon doing presentment (Validations) and it's mandatory for URA. **NB: Only for URA requests**	Optional	String
Location	For NWSC payments indication meter number location e.g. Kampala, Jinja, Entebe etc.	Optional	String
processid	Generated and returned during UEDCL presentment. It should be supplied for both amount validation and payment requests **NB: Only for UEDCL requests**	Optional	String
uedlid	Returned by UEDCL on both operations for presentment and amount validation. **NB: Only for UEDCL requests**	Optional	String
customerMobile	Customer's phone number	Mandatory	String
ProductCode	To be supplied while making payment for DSTV and GOTV only. Returned upon fetching TV packages	Optional	String
Amount	Bill Amount to be paid	Mandatory	String
Operation	Specifies action to be taken e.g fetchtvPackages for DSTV/GOTV, amount validation for UEDCL e.t.c	Optional	String
TVcustomernumber	Returned on GoTV and DSTV Validation response and is to be supplied while fetching TV packages for DSTV or GoTV.	Optional	String
schoolCode	Returned on SchoolPay Validation response and is to be supplied for SchoolPay payments	Optional	String

Bill Payment Response Messages Fields

Parameter name	Description	Required	Format
StatusDesc	Transaction Status	Mandatory	String
requestReference	Original trans reference	Mandatory	String
FlexipayRefNum	Felxipay reference number	Mandatory	String
StatusCode	Transaction status code	Mandatory	String
Amount	Bill amount to be paid.	Optional	String
customer_name	Customer name tied to the customer id being verified	Mandatory	String
pending_balance	Amount due for the customer (0 if missing)	Mandatory	String
Payment_mod	Indicator if the customer can pay any amount (PART), exact amount (EXACT) or anything equal or higher than pending amount (FULL)	Mandatory	String
externalreference	Transaction ID for the consuming third party system	Mandatory	String
Tin	Taxpayer Identification Number. This is returned upon doing presentment (Validations) and it's mandatory for URA only.	Optional	String
processid	Generated and returned during UEDCL presentment. It should be supplied for both amount validation and payment requests	Optional	String
uedlid	Returned by UEDCL on both operations for presentment and amount validation.	Optional	String
defaultPackage	Default customer TV package, Returned by multichoice(DSTV/GOTV) **NB: Only for DSTV and GOTV requests**	Optional	String
TVcustomernumber	Multichoice customer number. Returned by multichoice(DSTV/GOTV) **NB: Only for DSTV and GOTV requests**	Optional	String
Tvpackages	All Packages returned under a specific provider, in this case DSTV and GOTV **NB:** productCode and packagename are separated with symbol ^ then packagename and packagePrice are separated with a pipe(|) **Example:(**productCode ^ packagename| packagePrice)	Optional	String
AvailableBal	Amount of money in the account that is available and can be utilized.	Optional	String

ActualBal	Amount of money that is in the account at any given time.	Optional	String
Token	Yaka token from Umeme	Optional	String
Units	Yaka units from Umene	Optional	String

Error Response

When the Authorization token is not passed in the headers or when it has expired.

{

"httpCode": "401",

"httpMessage": "Unauthorized",

"moreInformation": "Cannot pass the security checks that are required by the target API or operation, Enable debug headers for more details."

}

When your Signature doesn't match the request body sent.

{

"httpCode": 401,

"httpMessage": "Unauthorized",

"moreInformation": "Signature Validation Failed."

}

When you pass wrong X-IBM-Client-Id and Secret

{

"httpCode": "403",

"httpMessage": "Forbidden",

"moreInformation": "Multiple client ids found."

}

Response Codes

Response Code	Description
00	Success
01	Pending
57	Failed
51	Insufficient Funds
58	Invalid Client ID or Authentication failed
59	Invalid or empty payload
94	Duplicate transaction (RequestID)
99	Read timed out
37	Missing Transaction Reference
04	Invalid or amount not supplied
02	Missing customer phone number
16	Single transaction limit exceeded
19	Daily transaction account limit exceeded
20	Daily account balance limit exceeded
17	Amount below set limit
102	Debit account is missing